

CHE 101
PRACTICE
QUESTIONS
BY IFEKITAN

1. Q: What determines the degree of completeness of a reaction?

A: catalyst

B: intimacy of contact

C: rate of reaction

D: equilibrium constant

2. Q: What is the substance called that causes an alteration of the speed of a chemical reaction?

A: variable

B: gas

C: catalyst

D: equilibrium constant

3. Q: What substance does not have catalytic capability, but does increase the effectiveness of a catalyst?

A: enzyme

B: promoter

C: anti-catalyst

D: true constant

4. Q: Who first proposed the atomic theory?

A: John Daltan

B: Julius Robert

C: Lord Kelvin

D: William Thomson

5. Q: What is the temperature at which the vapor pressure of a liquid slightly exceeds the pressure of the atmosphere above the liquid?

A: absolute zero

B: boiling point

C: Kelvin

D: melting point

6. Q: When the pressure on a liquid is increased, the boiling point:

A: goes up

B: goes down

7. Q: What is the property of a fluid that prevents it from flowing when subjected to an applied force?

A: compressibility

B: volume

C: viscosity

D: density

8. Q: High-viscosity fluids tend to:

A: resist flow

B: flow easily

9. Q: What theory states that equal volumes of different gases contain the same number of molecules when compared under the same conditions.

A: Quantum Theory

B: Kinetic Theory

C: Molecular Law

D: Avogadro's Law

10. Q: What are atoms of the same element that differ in weight?

A: isotopes

B: enzymes

C: neutrons

D: ions

11. Q: Who first came up with the periodic table?

A: Amedeo Avogadro

B: Dmitry Mendeleev

C: Charles Darwin

D: Isaac Newton

12. Q: What is the number of the element on the Periodic Table called?

A: periodic number

B: element number

C: quantum number

D: atomic number

13. Q: Who invented the X-ray?

A: Wilhelm Conrad Roentgen

B: Marie Curie

C: Pierre Curie

D: Ernest Rutherford

14. Q: What is the center of the atom?

A: atom core

B: nucleus

C: electron

D: proton

15. Q: Who's theory proposed that electrons are arranged in shells, or quantum levels, at a distance from the nucleus?

A: Rutherford

B: Newton

C: Bohr

D: Davis

16. Q: If helium has an atomic number of 2, how many electrons does it have?

A: 1

B: 2

C: 3

D: 4

17. Q: According to Bohr's Theory, what is the maximum number of electron orbital layers?

A: 2

B: 3

C: 5

D: 7

18. Q: Which gas is not an inert gas?

A: oxygen

B: neon

C: argon

D: helium

19. Q: Inert gases have:

A: one electron in the outer shell

B: completely filled outer shells

C: two electrons in the outer shell

D: no electrons

20. Q: Do inert gases enter into chemical combinations in nature?

A: no

B: yes

21. Q: The inner-most electron shell can contain how many electrons?

A: 7

B: 4

C: 2

D: 1

22. Q: Which is a proton donor?

A: acid

B: base

23. Q: Which is a proton acceptor?

A: acid

B: base

24. Q: Alkali metals have:

A: low melting points

B: high melting points

25. Q: How many Alkali metals are there?

A: 2

B: 4

C: 5

D: 6

26. Q: Which is not an Alkali metal?

A: lithium

B: sodium

C: gold

D: potassium

27. Q: What was Melvin Calvin was known for his study of?

A: x-rays

B: photosynthesis

C: magnetism

D: boiling point

28. Q: What element is crucial to the existence of living organisms?

A: iron

B: lithium

C: carbon

D: neon

29. Q: What is the process of rapid oxidation of a substance with simultaneous evolution of heat?

A: viscosity

B: melting

C: explosion

D: combustion

30. Q: What is the gradual change of a liquid into a gas without boiling?

A: evaporation

B: melting

C: combustion

D: boiling

31. Q: What is the breaking down of a substance or compound into its simpler components?

A: combustion

B: decomposition

C: melting

D: combination

32. Q: What are chemical changes in organic substances caused by the action of enzymes called?

A: reaction

B: oxidation

C: enzymiphication

D: fermentation

33. Q: What compounds are formed by the reaction of acids and alcohols?

A: ethers

B: enzymes

C: esters

D: bases

34. Q: Which of the following elements must be present for fire to exist?

A: oxygen or chlorine

B: oxygen or carbon

C: carbon or lithium

D: neon or iron

35. Q: What is the number of electrons that an atom can give to or accept from another atom in a chemical reaction?

A: no more than 2

B: valence

C: atomic number

D: electron shell

36. Q: Who was key to developing the valence theory?

A: Melvin Calvin

B: Henry Dow

C: Sir Edward Franklin

D: Humphry David

37. Q: What is the temperature at which a liquid congeals into the solid state at a given pressure?

A: boiling point

B: melting point

C: solid point

D: freezing point

38. Q: What is the change of a substance from the solid to the liquid state called?

A: decomposition

B: fusion

C: combustion

D: explosion

39. Q: Who invented vulcanization for the rubber industry?

A: Charles Goodyear

B: James Firestone

C: Robert Michelin

D: Albert Brookstone

40. Q: What process combines rubber and sulfur at a high temperature?

A: fulmination

B: vulcanization

C: fermentation

D: distillation

41. Q: Who won the 1944 Nobel Prize in chemistry for his work in nuclear fission?

A: Fritz Strausmann

B: Joseph Lussac

C: Otto Hahn

D: Johann Glauber

42. Q: How many Halogens are there?

A: 8

B: 7

C: 6

D: 5

43. Q: What elements are Hydrocarbons composed of?

A: Hydrogen and Chlorine

B: Argon and Uranium

C: Hydrogen and Carbon

D: Oxygen and Carbon

44. Q: What is something called when it changes color in response to the nature of its chemical environment?

A: enzyme

B: indicator

C: promoter

D: changer

45. Q: What color does Litmus change to in acid solutions?

A: red

B: orange

C: blue

D: green

46. Q: What color does Litmus change to in basic solutions?

A: red

B: blue

C: orange

D: green

47. Q: What does an atom that loses an electron form?

A: cation

B: anion

48. Q: What does an atom that gains an electron form?

A: cation

B: anion

49. Q: What is one of two or more species of atom having an identical atomic number but differing in mass number called?

A: metal

B: enzyme

C: resin

D: isotope

50. Q: What is one of two or more molecules, having the same chemical compositions, but a differing arrangement of atoms?

A: isomer

B: isotope

C: ion

D: cation

51. Q: Who invented the Carbon-14 dating method?

A: Irving Langmuir

B: Williard Libby

C: August Hofmann

D: Otto Hahn

52. Q: Amalgams are types of what?

A: alloys

B: elements

C: metals

D: isotopes

53. Q: Alloys are:

A: pure elements

B: combinations of carbon and other elements

C: combinations of metals

D: always acids

54. Q: What is the ability of a solid substance to resist abrasion and surface deformation called?

A: hardness

B: softness

C: conductivity

D: tensile strength

55. Q: What is the resistance to breakage called?

A: hardness

B: elasticity

C: malleability

D: tensile strength

56. Q: What is the ability to return to the original shape after being deformed called?

A: hardness

B: elasticity

C: malleability

D: tensile strength

57. Q: What is the ability to be shaped called?

A: hardness

B: elasticity

C: malleability

D: tensile strength

58. Q: What is the sum of all the atomic weights of all the atoms in a molecule?

A: Electron Weight

B: Neutron Weight

C: Molecular Weight

D: Heavyness

59. Q: What is the smallest particle of a substance having the specific chemical properties of that substance?

A: atom

B: molecule

C: electron

D: nucleus

60. Q: What term indicates the hydrogen ion concentration of a solution?

A: dH

B: pD

C: Lh

D: pH

61. Q: What is the chemical formula for salt?

A: NaCl

B: H₂O

C: CaSO₄

D: MgBr₂

62. Q: What is the chemical formula for water?

A: NaCl

B: H₂O

C: WaTr

D: MgBr₂

63. Q: What is a solvent for salt?

A: oil

B: carbon

C: iron

D: water

64. Q: Polymers are made up of smaller repeating units called?

A: strings

B: cells

C: monomers

D: elements

IFEKITAN CARES

Solutions

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